Automating Passive Web Application Recon With n8n & AI



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ABOUT CW LABS



CW Labs is a renowned UK based Ed-tech company specializing in cybersecurity cyber range labs. They provide on-demand educational services and recognize the need for continuous adaptation to evolving threats and client requirements. The company has these primary divisions:

1. Cyber Range Labs
2. Up-Skilling Platform
3. Infinity Learning Platform

CWL provides customized on-site training for organizations and accepts bulk orders for corporate upskilling. For inquiries or more details, feel free to reach out at **support@cyberwarfare.live.**

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ABOUT ME

Abhijeet Kumar

Security Researcher

- ★ Security researcher specializing in adversary simulations, executing multi-stage attack chains from initial foothold to data exfiltration.
- ★ Experienced in CI/CD pipelines, Kubernetes, Active Directory, cloud infrastructures (AWS/Azure/GCP), and *NIX systems.
- ★ Maintains a homelab to replicate adversary TTPs and analyze emerging threats.
- ★ Avid reader and home cook

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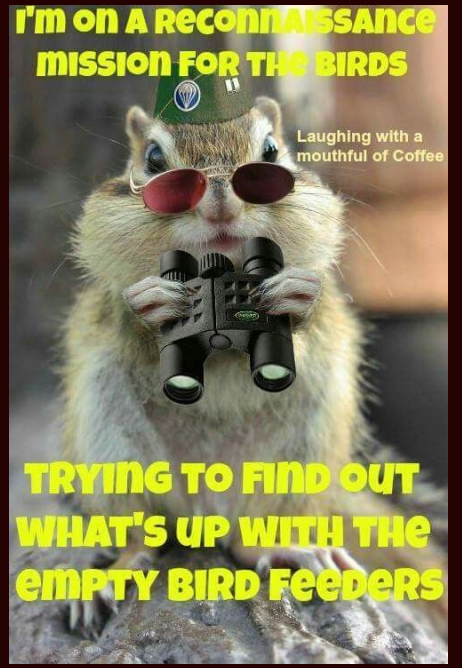
AGENDA

- ★ Passive Recon 101
- ★ Red Team Goals
- ★ Intro to n8n
- ★ Workflow Overview
- ★ Something Something
- ★ Workflow Limitations

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PASSIVE RECON 101

- ★ Identify attack surface without touching the target.
- ★ Avoid triggering security monitoring systems.
- ★ Correlate publicly available reconnaissance data.
- ★ Reduce unnecessary active enumeration.



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RED TEAM GOALS

- ★ Enumerate externally exposed application assets.
- ★ Discover hidden application functionality.
- ★ Fingerprint technologies and infrastructure.
- ★ Prioritize high-value attack vectors.
- ★ Build reconnaissance before exploitation.



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INTRO TO N8N

- ★ n8n is a workflow orchestration platform that automates repetitive reconnaissance tasks by integrating multiple OSINT tools, APIs, and AI models into a single pipeline.
- ★ Core capabilities include :-
  - API Integration
  - Workflow Automation
  - Data Correlation
  - AI Integration
  - Report Generation



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- Typical use cases include :-
  - Eliminate repetitive manual tasks
  - Standardize reconnaissance workflows
  - Scale intelligence collection
  - Build reusable recon pipelines

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WORKFLOW OVERVIEW



Normalize target

STEP 1

Convert the target into a standard structure so domains, brands, handles, and identifiers can be processed consistently.



Collect passive intel

STEP 2

Gather reconnaissance from diverse passive sources without directly touching the target infrastructure.



Parse and enrich

STEP 3

Extract useful entities, interpret raw findings, and attach context that makes signals more actionable.



Correlate duplicates

STEP 4

Automatically merge overlapping observations to reduce noise and preserve the strongest evidence trail.



Generate AI reports

STEP 5

Transform the correlated dataset into concise reconnaissance reports with structured, analyst-ready takeaways.

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WORKFLOW LIMITATIONS

- ★ Passive data may be outdated.
- ★ Coverage depends on public sources.
- ★ Hidden assets remain undiscovered in most instances.
- ★ Findings requires manual validation.
- ★ Reconnaissance does not prove vulnerabilities.

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